Free and Open Source Tools for Research

Jesse Oldroyd

June 15, 2026

What is FOSS?

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Free and open source software (FOSS) is software

- that is freely available (usually under a permissive license);
- has accessible source code.

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Philosophy of Free Software

The philosophy of free software can be encapsulated by *four essential freedoms* [2]:

1. to use the software.
2. to inspect and change the software.
3. to share copies of the original software.
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The freedoms associated with the use of free software are given legal weight by various licenses, classified as *copyleft* or *permissive* [3].

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When FOSS Is the Wrong Choice

Despite good reasons to use them, free and open source tools are not always a good idea. You need to be mindful of the following:

- Use tools that work well with what your collaborators are using.
- FOSS tools are often not as polished as their proprietary counterparts.
- Security concerns.

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Tools

Types of Tools

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- One of the most fundamental tools is a text editor.

- Even though you start with text, output is not necessarily just a text file!

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Tools for Computation

- R
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- Python with NumPy
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Python Code for Statistics

```python
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
sns.set_theme(style="dark")

# Simulate data from a bivariate Gaussian
n = 10000
mean = [0, 0]
cov = [(2, .4), (.4, .2)]
rng = np.random.RandomState(0)
x, y = rng.multivariate_normal(mean, cov, n).T

# Draw a combo histogram and scatterplot with density contours
f, ax = plt.subplots(figsize=(6, 6))
sns.scatterplot(x=x, y=y, s=5, color=".15")
sns.histplot(x=x, y=y, bins=50, pthresh=.1, cmap="mako")
sns.kdeplot(x=x, y=y, levels=5, color="w", linewidths=1)
```

Listing 1: Example Python code for statistical visualization

Visualization

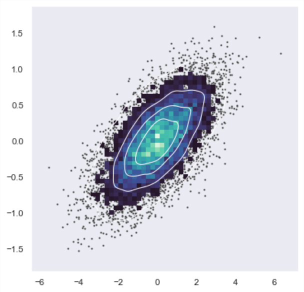

Figure 1: Data visualization produced by Listing 1

IDE vs. Text Editor

- When using programming languages, you will need a program to interact with the language.

- There is often a choice between an *integrated development environment* (IDE) and a text editor.

- Open source IDEs: RStudio, Spyder, Octave (GUI).

- Open source text editors: VSCode (new school), Vi(m) and Emacs (*very* old school).

General or specific language support

IDEs are tailored to specific languages (such as RStudio). Text editors tend to be more multi-purpose.

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Tools for Drawing

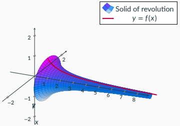

Figure 2: A PGFPlots drawing

- matplotlib
- pandas
- TikZ/PGFPlots
- Inkscape

Composability Again

- The tools mentioned so far are not mutually exclusive!

- These tools have specific purposes, but can be chained together to form a powerful workflow.

Solving a Math Problem

The expanded form of $(x - 3)^5$ is $x^5 - 15x^4 + 90x^3 - 270x^2 + 405x - 243$.

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References

[1] *Open Source - GNU Project - Free Software Foundation (FSF).* July 3, 2007. URL: `https://web.archive.org/web/20070703091340/https://www.gnu.org.ua/philosophy/free-software-for-freedom.html` (visited on 06/14/2026).

[2] *Philosophy of the GNU Project - GNU Project - Free Software Foundation*. URL: `https://www.gnu.org/philosophy/philosophy.html` (visited on 06/14/2026).

[3] *Choose a License – Open Source Programs Office*. URL: `https://ospo.library.jhu.edu/learn-grow/licensing-overview/choose-a-license/` (visited on 06/14/2026).